

Designer Blouse AI Platform

Implementation Planning & Execution Requirements

Document Version	1.0.0
Based On	Approved PRD v1.1.0
Target Market	India — Mobile-first
Audience	AI coding agents and engineering teams
Delivery Goal	Working, tested, deployable V1 website

Purpose: give an AI coding agent an ordered, verifiable plan to build the V1 website from repository setup through deployment. This is an execution guide, not a replacement for the approved product requirements.

1. Mission & Instructions to the AI Agent

Build the Designer Blouse AI Platform as a production-minded V1 application. Follow the approved PRD for user-facing scope and behaviour. Work in small phases, validate each phase before continuing, and keep the application runnable throughout development.

- Inspect the existing repository before changing files. If empty, initialize the project using the approved stack.
- Treat the approved PRD as the source of truth for scope, fields, workflows, and acceptance criteria.
- Do not implement V2 cart, checkout, payments, inventory reservation, shipping, or 3D viewer features in V1.
- Never hard-code secrets, business phone numbers, storage credentials, or provider API keys in client code.
- Use typed data contracts, reusable components, accessible UI patterns, and documented environment variables.
- Do not claim a feature is complete unless implemented and its relevant checks have run.
- If third-party credentials are unavailable, create a documented adapter boundary and local/test mode; do not present mock output as real production AI output.
- After each phase, run linting, type checks, tests, and production build as applicable; report failures and unresolved assumptions.

2. Delivery Definition

The project is complete when customers can browse fabrics, match saree colors, create a three-view AI blouse design, open a public permalink, and start an enquiry through WhatsApp. Admins can manage catalog products and inspect AI usage. Commerce remains disabled.

Workstream	Completion Outcome
Catalog	Responsive catalog, filters, product details, imagery, availability and pricing.
Color matching	Image upload, loupe, shade adjustment, Hex-to-CIELAB conversion and ranked results.
AI Studio	Validated upload, style selection, asynchronous job lifecycle, three views and quota enforcement.
Sharing & conversion	Public design page and dynamically encoded WhatsApp enquiry link.
Admin	Protected product management, color calibration, generation audit and operational visibility.
Quality & deployment	Security policies, responsive QA, environment setup, deployed app and acceptance checklist.

3. Recommended Technical Decisions

Area	Implementation Direction
Application	Next.js App Router, TypeScript, Tailwind CSS; responsive and accessible UI.
Data & auth	Supabase PostgreSQL and Supabase Auth; server-side authorization for privileged operations.
Storage	Cloudflare R2 for catalog imagery, uploads and generated assets; controlled upload and access policies.
Background processing	Upstash QStash / Redis or equivalent queue and webhook flow for long-running AI tasks.
AI provider	Fal.ai or Replicate behind a provider adapter; configurable provider/model.
Color utilities	Tested sRGB → CIEXYZ (D65) → CIELAB conversion and CIE76 ΔE.
Messaging	WhatsApp Click-to-Chat URL generated from validated product/design data.
Hosting	Deploy to a suitable Next.js host; document free-tier limits and required paid upgrades.

Free-tier availability, vendor pricing and service limits can change. Verify current limits before launch and do not depend on unverified quotas.

4. Execution Phases & Exit Gates

Phase	Tasks	Exit Gate
0 — Discovery	Inspect repository, record assumptions, map routes, entities and required accounts.	Plan and configuration checklist documented.
1 — Foundation	Initialize app, linting, type checks, design tokens, layout, navigation and env validation.	App runs locally; production build passes.
2 — Data & security	Schema, migrations, indexes, color RPC, RLS, auth and storage strategy.	Migrations apply; access-control tests pass.
3 — Catalog & admin	Catalog grid, filters, product details, image gallery and protected product CRUD.	Seed products display; mutations are admin-protected.
4 — Color matching	Upload, loupe, eyedropper, tuning controls, conversion, RPC integration and result grid.	Color test cases pass; results ordered by ΔE .
5 — AI Studio	Upload validation, style controls, quota, provider adapter, job callbacks, retries and output storage.	Test job reaches auditable completed/failed state.
6 — Sharing & WhatsApp	Public /d/:id page, zoomable views, style summary and encoded enquiry CTA.	Permalink works without login; CTA is valid.
7 — Hardening	Usage dashboard, responsive QA, accessibility, security and performance checks.	Acceptance criteria recorded; critical defects resolved.
8 — Deployment	Secrets, migrations, domain, storage CORS, callbacks, monitoring and runbook.	Deployed smoke tests pass; handover complete.

5. Application Routes & Screens

Route	Requirements
/	Landing page with catalog, color match, AI Studio entry points and featured fabrics.
/catalog	Product grid; fabric type, occasion and price filters; responsive cards and empty/loading/error states.
/products/[slug-or-id]	Product imagery, SKU, title, details, price/unit, availability, swatches, studio entry and WhatsApp CTA.
/color-match	Saree upload, sampling, tuning sliders, Hex output, match results and reset/retry controls.
/studio	Fabric upload, validation, optional product association, style selectors, job status and output views.
/d/[id]	Public design permalink with reference image, three views, style tags, target swatch, zoom and WhatsApp CTA.
/admin	Protected dashboard with generation totals, failure rates, latency and popular styles.
/admin/products	Protected product CRUD, imagery, eyedropper, color values, availability and pricing.
/auth/*	Sign-in and account flows required for authenticated quotas and admin access.

6. Database & Data Integrity

- Create versioned migrations for profiles, products, AI designs and AI usage logs as defined in the PRD.
- Use UUIDs, timestamps, unique SKUs, valid unit types/statuses, foreign keys and indexes for catalog filters, availability, design lookup and quota checks.
- Implement color proximity search using CIE76 ΔE ; return available products ordered by distance and filtered by the configured threshold.
- Store image URLs and provider job IDs; do not store large image binaries directly in PostgreSQL.
- Enforce the rolling 24-hour per-user quota atomically on the server to prevent concurrent requests from exceeding limits.
- Define retention/deletion behaviour for profiles, uploads, failed jobs and public design pages.
- Seed demo products for local testing and clearly distinguish demo content from production catalog data.

6.1 Security & Authorization

- Enable row-level security on tables exposed through client-accessible APIs.
- Public users may read only public products and completed design records explicitly intended for sharing.
- Only authorized admins may manage products or inspect privileged audit data; sign-in alone does not imply admin access.
- Service-role keys and AI provider credentials must remain server-side.
- Validate ownership and permissions in server actions, API routes, queue callbacks and storage operations.
- Verify callback authenticity where supported and make job completion handlers idempotent.

7. Frontend Implementation Requirements

7.1 Design System

- Create consistent spacing, typography, buttons, forms, cards, badges, dialogs, toasts and loading states.
- Use mobile-first layouts, touch-friendly controls, responsive images and sticky CTAs only when they do not obstruct content.
- Include skeletons, empty/error states, retry actions and clear success feedback.
- Support keyboard navigation, visible focus, meaningful labels, alt text and sufficient contrast.
- Optimize images and provide a usable zoom interaction for generated views.

7.2 Catalog & Product Details

- Implement server-backed product retrieval, filters, pagination or incremental loading, and explicit unavailable states.
- Format prices for India and show the correct unit: meter, piece or yard.
- Keep product title, SKU, price, availability and color data consistent across catalog, detail, studio and WhatsApp payload.
- Validate filter query parameters and preserve expected browser back/refresh behaviour.

8. Color Matching Implementation Plan

- Build a preview that accounts for image orientation and scaling so sampled coordinates correspond to the displayed image.
- Implement a circular magnifier at approximately 2.5× zoom with a central crosshair; support touch and pointer input.
- Extract selected pixels and update Hex/swatches immediately.
- Implement lightness and saturation tuning of approximately ±20%, with visible reset-to-original behaviour.
- Unit-test sRGB gamma linearization, D65 CIEXYZ conversion and CIELAB conversion against known reference cases.
- Call the database matching function with sampled coordinates; display ΔE and sort ascending.
- Explain that camera processing, lighting, display calibration and dye-lot variation make digital matching an estimate, not a physical guarantee.

9. AI Studio & Background Job Plan

State	Expected Behaviour
Pending	Request is validated, persisted and queued.
Processing	Worker submits provider job(s); UI displays progress and prevents duplicate submissions.
Completed	All three images are stored, linked to the design and available on the permalink.
Failed	Failure is recorded with a safe user-facing message and protected diagnostic details.

- Validate extension, MIME type, actual image decoding, file size, dimensions and upload permissions.
- Normalize orientation and use safe storage keys; never trust client-provided filenames.
- Create a design record before dispatching generation; track provider job IDs and per-operation status.
- Use callbacks or polling with timeouts, bounded retries, idempotency and cleanup of abandoned jobs.

- Mark a design complete only when front, back and 3/4 outputs are available.
- Use the front render as conditioning input for subsequent views when supported by the selected provider.
- Keep target Hex, style attributes, reference fabric and provider settings consistent across views.
- Enforce no more than 3 complete design requests per authenticated user per rolling 24 hours; explicitly decide and document whether failed attempts count.
- Provide a local test mode that simulates lifecycle states without misrepresenting mock images as real AI output.

10. WhatsApp Conversion Plan

- Store the business WhatsApp number in configuration and validate it before building links.
- Build enquiry content from trusted product/design data rather than arbitrary unvalidated client input.
- Include product title, SKU, displayed price/unit, matched Hex and ΔE when relevant, style attributes and public design URL.
- Encode the full message as a query parameter; test newlines, special characters, the rupee symbol and non-ASCII text.
- Track CTA clicks without collecting unnecessary personal information.
- Keep availability, sizing, tailoring timeline, payment and shipping confirmation manual in V1.

11. Admin & Operations Dashboard

- Implement explicit admin authorization and protected routes/actions.
- Support product create/read/update, availability toggles, image gallery management, color-family tagging and unit pricing.
- Provide admin eyedropper sampling and calculate CIELAB from the selected Hex value.
- Show generation counts, completed/failed rates, latency, provider, recent errors and popular style attributes.
- Allow safe inspection of job state and error details without exposing credentials or private user data.

12. Testing Strategy & Quality Gates

Test Type	Minimum Coverage
Unit tests	Color conversions, CIE76, Hex validation, URL encoding, quota logic, upload validation and feature flags.
Database tests	Migrations, RPC ranking/thresholds, constraints, RLS and authorization boundaries.
Integration tests	Product creation, storage uploads, queue dispatch/callback, status changes and image URL persistence.
End-to-end tests	Browse/filter catalog, color match, create design, open permalink and initiate WhatsApp enquiry.
Responsive QA	iOS Safari and Android Chrome at 360, 375, 390, 414 and 430 px, plus desktop baseline.
Failure-path tests	Invalid/oversized uploads, missing credentials, provider timeout, partial generation, retries and unavailable products.
Security checks	Unauthenticated admin access, cross-user access, quota bypass, forged callbacks, secret leakage and unsafe uploads.

12.1 Release Gate

- Lint, type checks, tests and production build pass.
- No known critical or high-severity security issue remains open.
- Every PRD launch acceptance criterion has a recorded pass/fail result.
- Commerce flags are off and cart, payment, checkout and shipping UI are not reachable.
- Environment variables, provider callback URLs and manual setup steps are documented.
- Deployed smoke tests confirm catalog, color matching, AI job flow, permalink and WhatsApp CTA.

13. Environment & Deployment Checklist

- Maintain separate local, preview/staging and production configuration.
- Configure Supabase URL, public key, server-only service role secret, migrations and auth redirect URLs.
- Configure R2 bucket, credentials, asset visibility, CORS, upload limits and lifecycle policies.
- Configure queue credentials, webhook verification, callback URLs and retry policy.
- Configure AI provider key, model identifier, provider limits and billing alerts.
- Configure WhatsApp business number and public site base URL.
- Keep commerce, payment and shipping feature flags disabled for V1.
- Set up error monitoring, generation-failure alerts, uptime checks and cost monitoring.
- Verify domain, HTTPS, canonical metadata, sitemap/robots strategy and permalink indexing behaviour.
- Document rollback, migration recovery, storage cleanup and provider outage procedures.

14. Cost & Operational Guardrails

Use the PRD's cost estimates as planning assumptions, not guaranteed provider pricing. Verify current vendor pricing and free-tier limits before deployment.

- Record generation cost and operation count per design.
- Bound retries and prevent duplicate generation from repeated taps or webhook retries.
- Configure spending alerts or hard limits where supported.
- Compress and size images appropriately without materially degrading visual quality.
- Define retention for abandoned uploads and failed temporary assets.
- Warn the project owner before enabling paid services, changing tiers or activating commerce.

15. Required AI Agent Handover

- Summary of delivered functionality and known limitations.
- Repository structure and key architectural decisions.
- Database migration and seed-data instructions.
- Environment variable reference with placeholder values only.
- Local setup, test, build and deployment instructions.
- External accounts, credentials, callback URLs and manual setup steps still required.
- Test results with pass/fail status and skipped tests.
- PRD acceptance checklist with evidence or test references.
- Known issues, cost assumptions, monitoring recommendations and V2 next steps.

16. Definition of Done

Do not mark the project done merely because pages render. V1 is done only when critical journeys work end to end, access controls are enforced, AI jobs have observable lifecycle states, color matching is tested, WhatsApp links are valid, deployed smoke tests pass, and all V1 commerce features remain disabled.

End of AI Agent Website Build & Implementation Plan • Version 1.0.0